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Data Preprocessing and Visualization

AIN2IL | MTD.ba SS2026

Hagenberg 22.04.2026

HAGENBERG | LINZ | STEYR | WELS



UNIVERSITY
OF APPLIED SCIENCES
UPPER AUSTRIA

What a Data Set Ideally Looks Like

0.99516	0.890813	0.933726	0.793397	0.826405	0.236946	-1
0.853206	0.611647	0.317486	0.633609	0.411492	0.985231	+1
0.387494	0.459847	0.815049	0.394526	0.678227	0.031886	-1
0.733515	0.640438	1.19068	0.639685	0.0793674	0.160503	+1
0.274817	0.261054	1.20056	0.689895	0.401913	0.277955	-1
0.329943	0.241299	0.848705	0.721673	0.973852	0.795238	-1
0.334784	0.350487	0.315131	0.928277	0.816343	0.558292	-1
0.481578	0.738839	0.0925513	0.294667	0.612725	0.573062	-1
0.0940846	0.278992	0.451819	0.900141	0.220497	0.541176	+1
0.421025	0.785714	0.449038	0.920612	0.420418	0.749187	-1
0.939446	0.0468747	0.15846	0.625944	0.198894	0.176125	+1
0.845362	0.767883	0.824993	0.725803	0.808218	0.63495	-1
0.484793	0.129329	0.0783719	0.465347	0.291457	0.254278	+1
0.399041	0.751829	0.763511	0.894785	0.47902	0.15156	-1
0.643232	0.615629	0.430261	0.0458972	0.446513	0.844081	+1
...	...	...	...	...	...	...

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...	...	...	...	...	...	...

...but what is reality really like?

Basic Setup

$$\begin{pmatrix} x_1^1 & \dots & x_d^1 \\ \vdots & \ddots & \vdots \\ x_1^l & \dots & x_d^l \end{pmatrix}$$

Let us assume that rows in such a matrix correspond to *samples* (l in total) and that columns correspond to *features/variables* (d in total). The entry x_j^i (the element in the i -th row and j -th column) corresponds to which value the feature j has for the i -th sample.

If a column consists a finite set of labels, we speak of a *categorical variable/feature*.

If a column contains numerical values/numbers, we speak of a *numerical variable/feature*.

In Reality ...

- We might have to **filter/remove samples** before analyzing/modeling data.
- We might have to **filter/remove features** before analyzing/modeling data.
- We might have to **compute/construct some features** before analyzing/modeling data.
- We might have to **transform features** into a format that is suitable for analyzing/modeling data.
- We might have to **deal with missing values**.
- We should use **basic analyses and visualization** to get a feeling about possible data quality issues and also to obtain some hints about basic patterns in the data.

In Reality ...

- Finding and implementing the right data cleansing and preprocessing pipeline can be a **tedious and time-consuming process**.
- It usually requires a lot of **experience, intuition, and domain knowledge**.
- Data cleansing and preprocessing can take **up to 80% of the overall effort** of a data science/AI project.
- As said, **exploratory visualizations and statistics** can help with finding the right preprocessing and getting first hints about patterns in the data.

Basic Statistics (column-wise)

- Number/percentage of missing values
- For numerical columns: min, max, mean, median, variance/standard deviation, quartiles
- For categorical columns: numbers/percentages of different categories

Basic Visualization

Single categorical feature: bar chart (recommended) or pie chart (not recommended)

Single numerical feature: histogram

Numerical feature vs. categorical feature: set of histograms (possibly overlaid) or box(-and-whisker) plot

Two categorical features: heatmap or two-dimensional bar chart

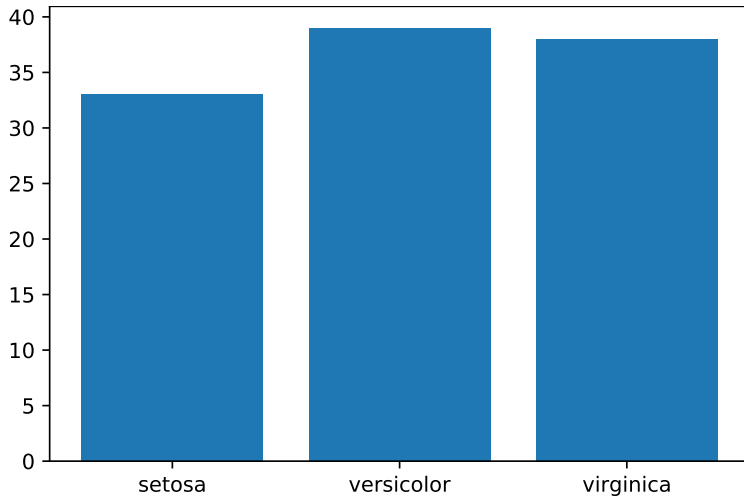
Two numerical features: scatter plot

Three numerical features: 3D scatter plot (hopefully rotatable)

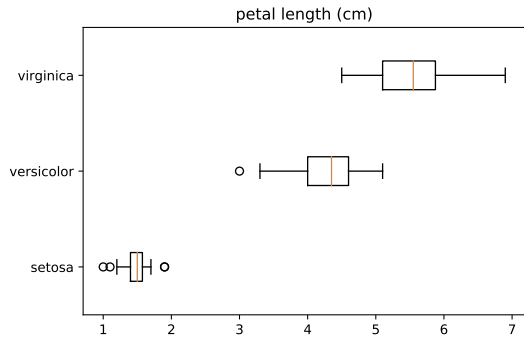
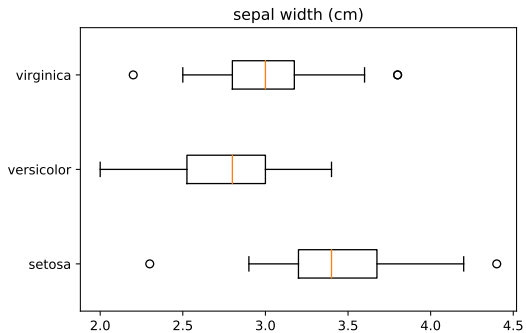
Two numerical features vs. categorical feature: color-labeled scatter plot

Three numerical features vs. categorical feature: color-labeled 3D scatter plot (hopefully rotatable)

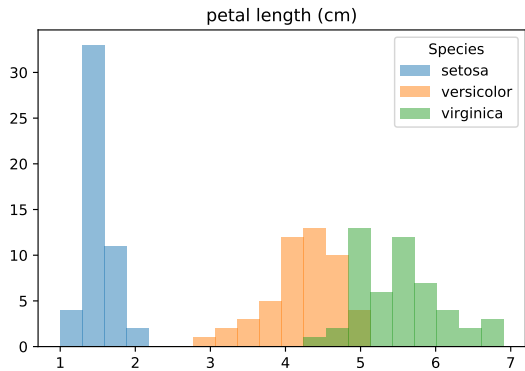
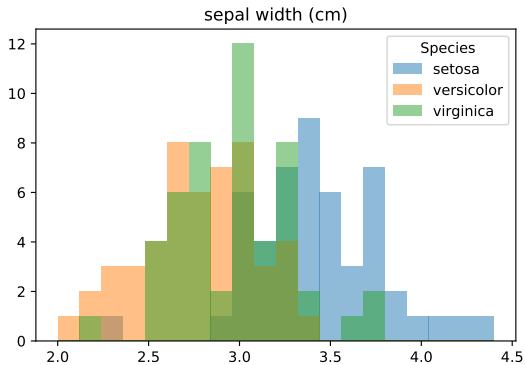
Bar Chart



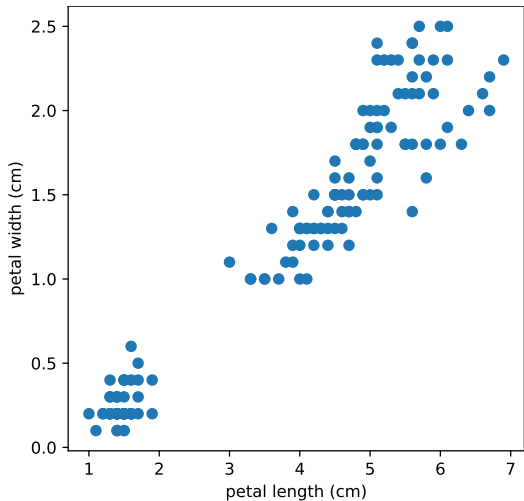
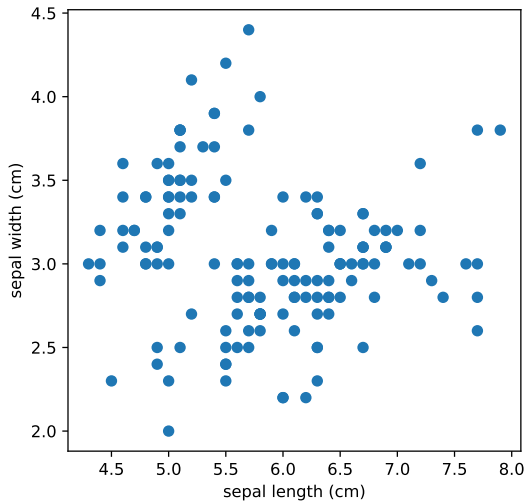
Box Plots



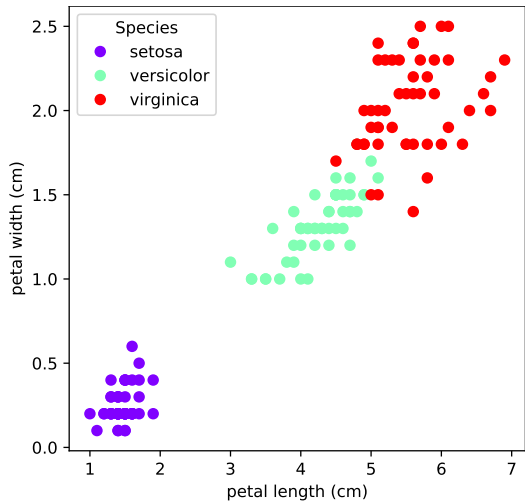
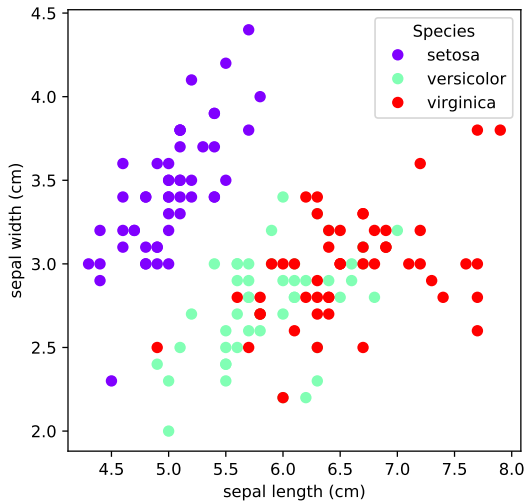
Labeled Histograms



Scatter Plots (unlabeled)



Scatter Plots (labeled)



Column Transformations

- Scale to **[0, 1]** or **[-1, +1]**
- Center and scale to mean 0 and variance 1
- Remove outliers (e.g. largest/smallest 1%, outside of interval of plausible values)
- Log-transform

Feature Construction

- Construct new features explicitly; example: suppose you have two columns "Price" and "Costs", then you might want to add two features as follows:

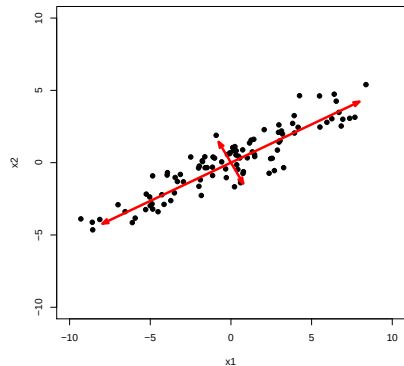
$$\text{Profit_abs} = \text{Price} - \text{Costs}$$

$$\text{Profit_rel} = \text{Profit_abs} \cdot 100 / \text{Costs}$$

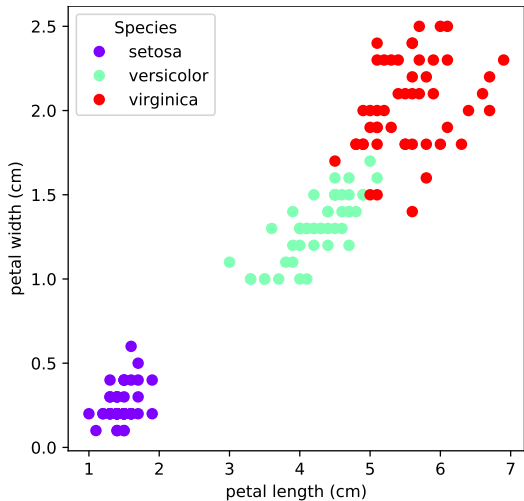
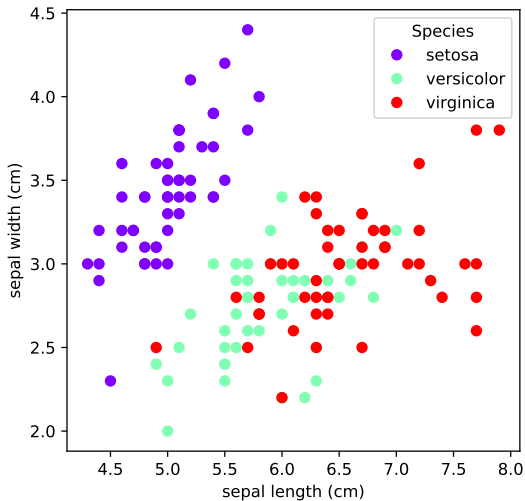
- Principal Component Analysis (PCA): computes rotated version of data for possible dimension reduction

Principal Component Analysis (PCA)

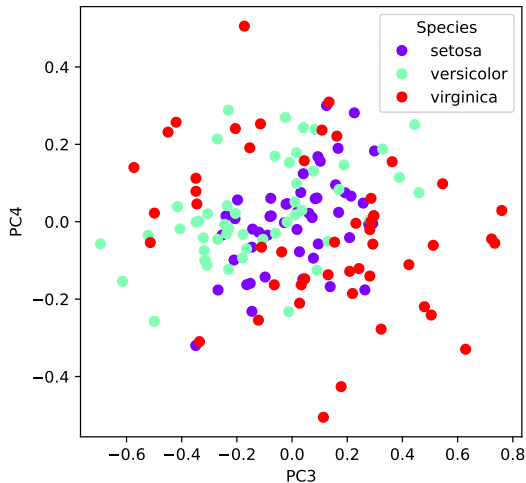
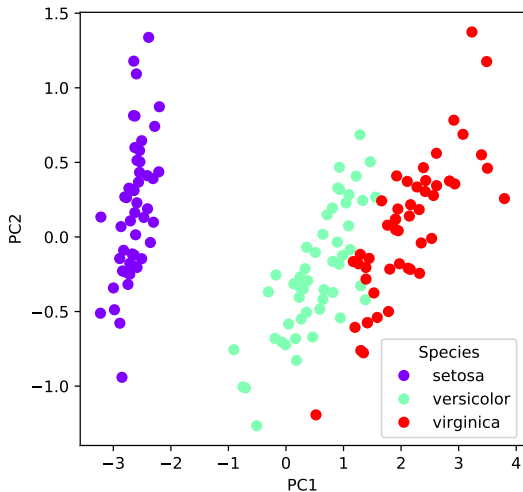
- Rotates centered (normal) vectorial data into linearly uncorrelated variables called *principal components*.
- This is accomplished by an orthogonal transformation that is typically determined using *singular value decomposition*.
- These projections are mutually uncorrelated, centered, and ordered descendingly according to variance.
- PCA can also be used for data dimensionality reduction by omitting all principle components that show low variance.



Scatter Plots (before PCA labeled)

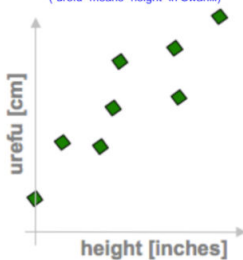


Scatter Plots (PCA, labeled)

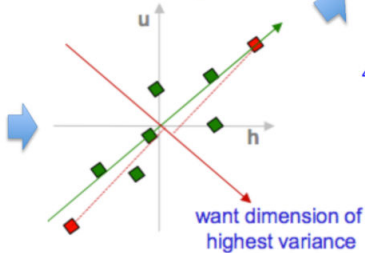


PCA in a nutshell

1. correlated hi-d data
("urefu" means "height" in Swahili)



2. center the points



3. compute covariance matrix

$$\begin{matrix} & h & u \\ h & \begin{pmatrix} 2.0 & 0.8 \end{pmatrix} \\ u & \begin{pmatrix} 0.8 & 0.6 \end{pmatrix} \end{matrix} \rightarrow \text{cov}(h, u) = \frac{1}{n} \sum_{i=1}^n h_i u_i$$

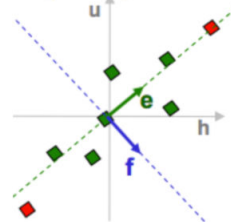
4. eigenvectors + eigenvalues

$$\begin{pmatrix} 2.0 & 0.8 \\ 0.8 & 0.6 \end{pmatrix} \begin{bmatrix} e_h \\ e_u \end{bmatrix} = \lambda_e \begin{bmatrix} e_h \\ e_u \end{bmatrix}$$

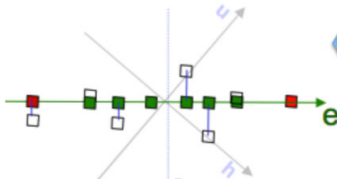
$$\begin{pmatrix} 2.0 & 0.8 \\ 0.8 & 0.6 \end{pmatrix} \begin{bmatrix} f_h \\ f_u \end{bmatrix} = \lambda_f \begin{bmatrix} f_h \\ f_u \end{bmatrix}$$

$\text{eig}(\text{cov}(\text{data}))$

5. pick $m < d$ eigenvectors
w. highest eigenvalues



7. uncorrelated low-d data



6. project data points to those eigenvectors

$$x'_e = x^T e = \sum_{j=1}^m x_j e_j$$

Recoding an Ordinal Feature as Numerical

Ordinal Feature = *categorical feature*, with the special property that the *categories are in a particular order*.

Example: school grade A > B > C > D > F (or SGT1 > GUT2 > BEF3 > GEN4 > NGD5)

Grade		Grade_num
A		1
A		1
B		2
A	→	1
F		5
C		3
D		4
A		1
B		2

Recoding an Categorical Feature as Numerical

If the labels of a categorical feature are not in a specific order and no two classes are more similar or more related to each other, the so called *one-hot encoding* is advisable.

Species	Species.setosa	Species.versicolor	Species.virginica
setosa	1	0	0
virginica	0	0	1
virginica	0	0	1
setosa	1	0	0
virginica	0	0	1
setosa	1	0	0
versicolor	0	1	0
versicolor	0	1	0
versicolor	0	1	0

Treatment of Missing Values

Options:

- Remove all samples with missing values
- Remove all features with missing values
- Imputation

Warnings/caveats:

- Removing leads to loss of data
- Both removing and imputing can introduce biases in the data, in particular, if the data are not randomly missing.

Imputation of Missing Data (numerical)

Options:

- Replace with fixed value (e.g. a price is missing if nothing has been ordered $\rightarrow 0$)
- Replace with mean or median
- Impute using a method that predicts one column from the others (see methods later)

Imputation of Missing Data (categorical)

Options:

- Replace with fixed value (e.g. patients are healthy if no disease is present)
- Replace with most frequent value
- Add missing as additional category (only meaningful if data are not missing randomly)
- Impute using a method that predicts one column from the others (see methods later)